CS & IT ENGINEERING

Data Structure & Programming

Trees and Graphs

Lecture No.- 08

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Recap of Previous Lecture

- Rotations
  - LL Rotation
  - RR Rotation

  Single Rotations (LL, RR)

  - LR Rotation
  - RL Rotation

  Double Rotations (LR, RL)

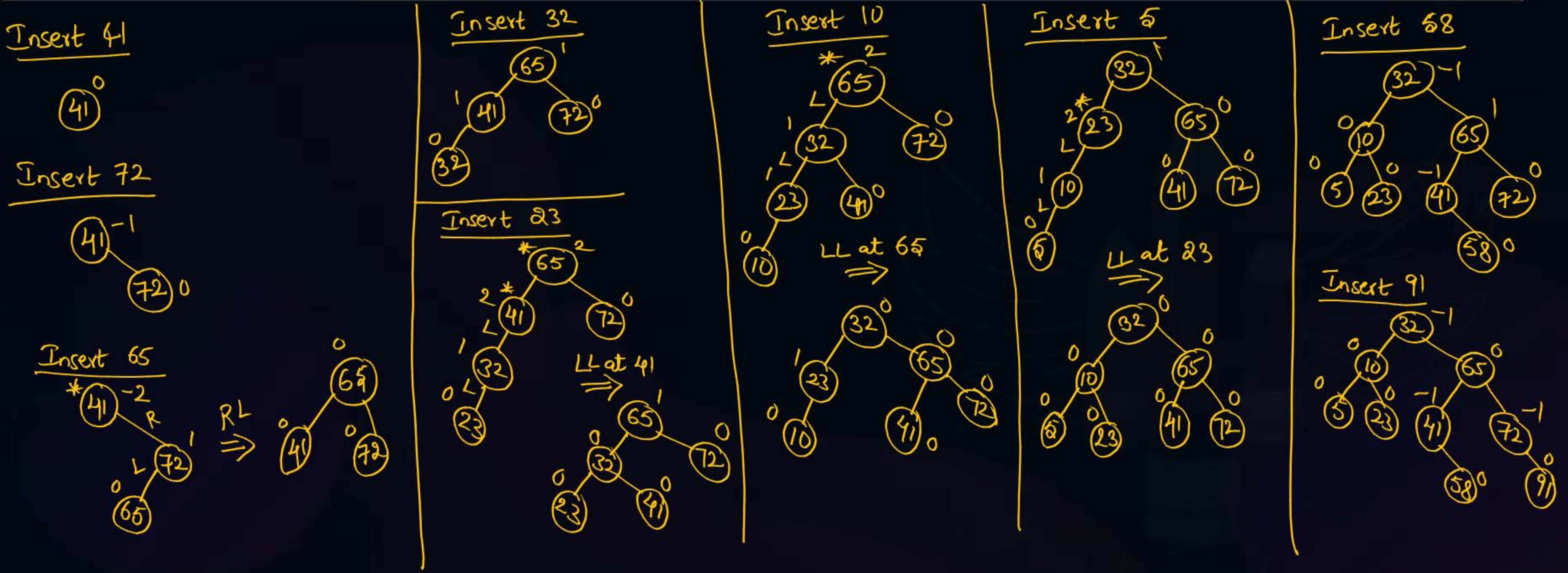
Topics to be Covered

- Homework Question Solving
- Graph Data structure
  - Nature
  - Types of Graphs
  - Graph Traversals

Topic : AVL Trees - 2

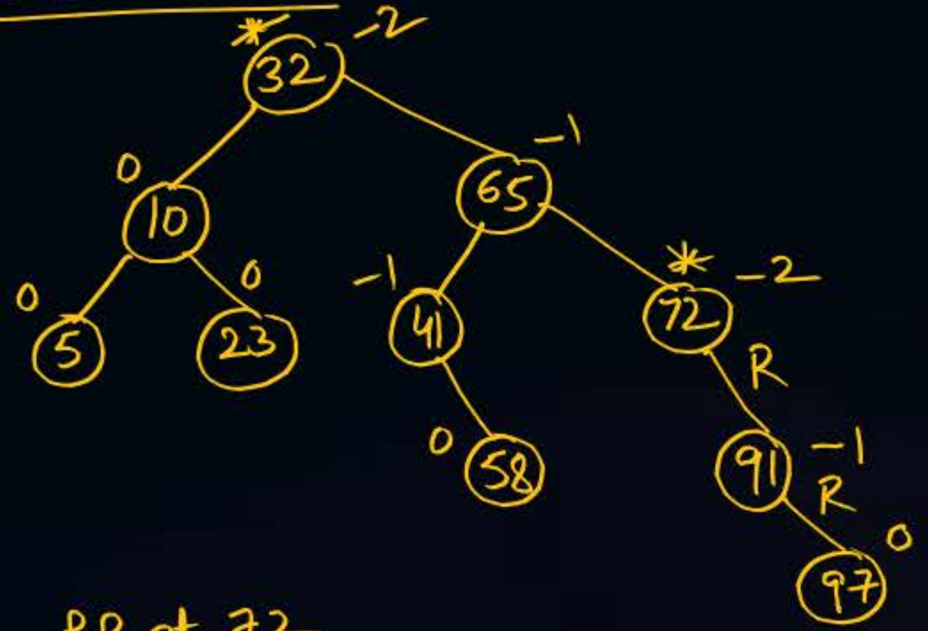
Home Work Question:

Construct an AVL Tree By Inserting Elements in the order: 41, 72, 65, 32, 23, 10, 5, 58, 91, 97, 99, 85, 70, 100

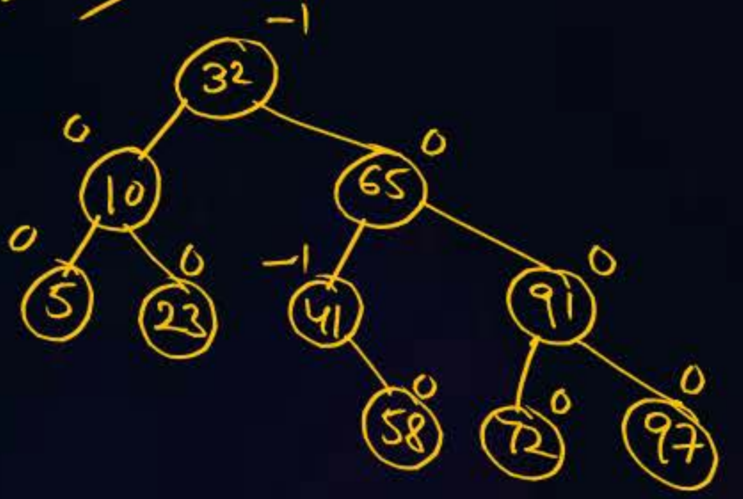


Topic : AVL Trees - 2

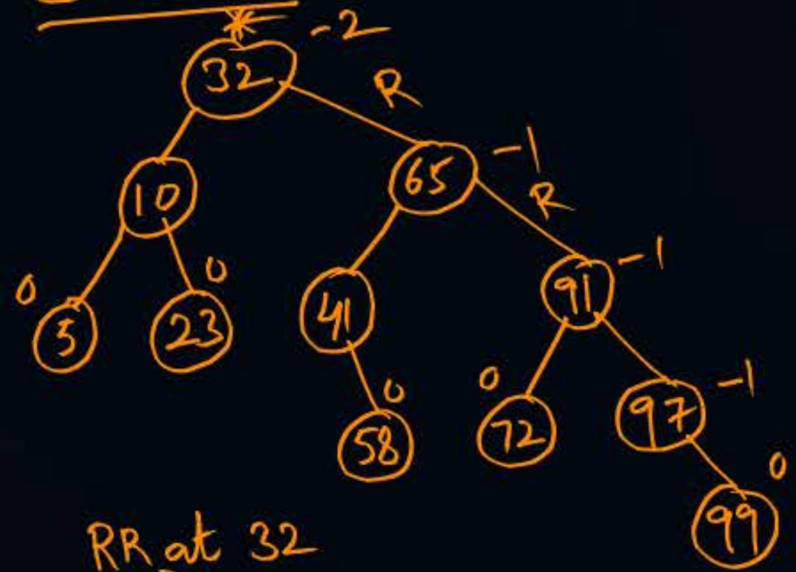
Insert 97



RR at 72 ⟹



Insert 99



RR at 32 ⟹

Insert ~~85~~ 70

Insert 100

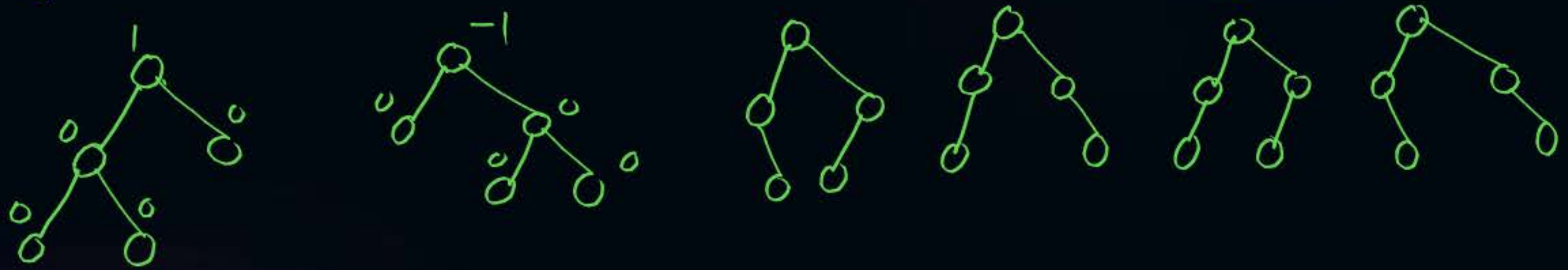
RR at 97 ⟹

Resultant AVL Tree

65, 32, 91, 10, 41, 72, 99, 5, 23, 58, 70, 85, 97, 100

- The minimum height of an AVL Tree Possible with 'N' nodes = $\left\lfloor \log_2 (N+1) \right\rfloor$

Let N = 5



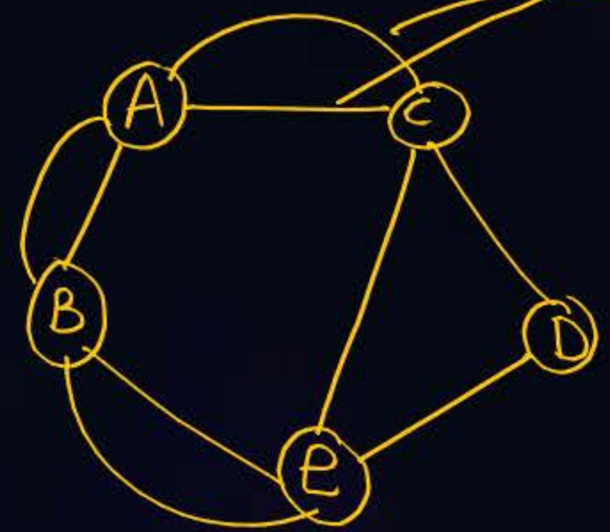
height = 2

- The Maximum height Possible for an AVL Tree with 'N' nodes = $1.44 * \log_2 N$

GRAPHS → DM – Types, Properties
→ DS – Traversals
→ Algorithms – Spanning Tree, shortest Path algo

- Non-linear Data Structure
- $G = \{V, E\}$
- It may be Cyclic (or) Acyclic

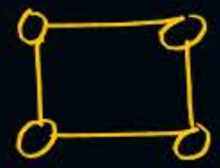
Ex:



Parallel edges

Types of Graphs

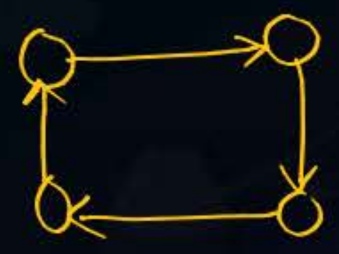
① Simple Graph : one Edge b/w vertices



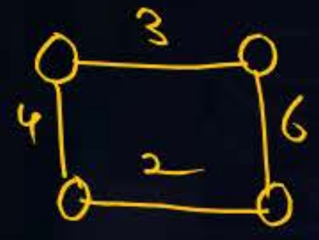
② Trivial Graph / Singleton Graph : only one vertex, No Edges

③ Isolated Graph : Multiple vertices, No Edges

④ Directed Graph : Edges are directed

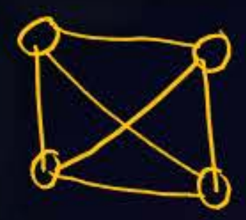


⑤ Weighted Graph : Edges are assigned with Weight | cost | distance



⑥ Complete Graph : if Every Pair of vertices are directly Connected to Each Other.

No. of edges $= {}^{N}C_2 = \frac{N(N-1)}{2}$



multigraph
Bipartite Graph
Dense Graph
Sparse Graph
⋮

Graph Traversals

- The sequence (or) order in which vertices are visited / accessed.

- 2 Types of Traversals:

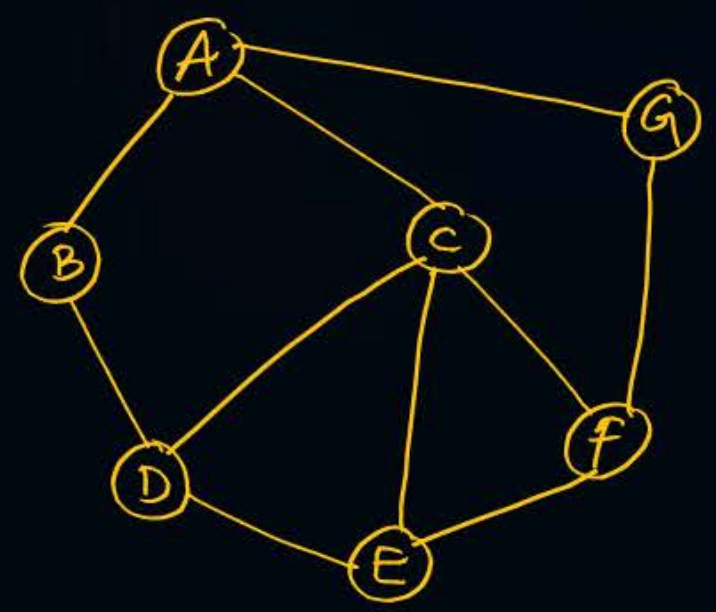
  1. Breadth First Traversal / Search : Queue DS
  2. Depth First Traversal / Search : Stack DS

- Exploration of a Vertex : when all immediate neighbour vertices of selected vertex 'X' are included, Then, 'X' is said to be Explored.

Breadth-First Traversal

- Select a vertex V, enqueue it into Q.
- Include all it's immediate neighbours in Q, [Non included] in any order.
- Once a vertex is Explored, Then Dequeue it from Q and output in the sequence.
- Repeat until all vertices are Explored.

Example



Valid BFS Sequences :

~~A~~	~~B~~	~~G~~	~~C~~	~~D~~	~~F~~	~~E~~

1. o/p: A B G C D F E
2. A C B G D E F
3. A B C G D F E

Invalid BFS Sequences

1. A G B C D F E Invalid BFS
2. A B G C F D E " "
3. A B D C G E F
4. A B G F C E D

Depth-First Traversal

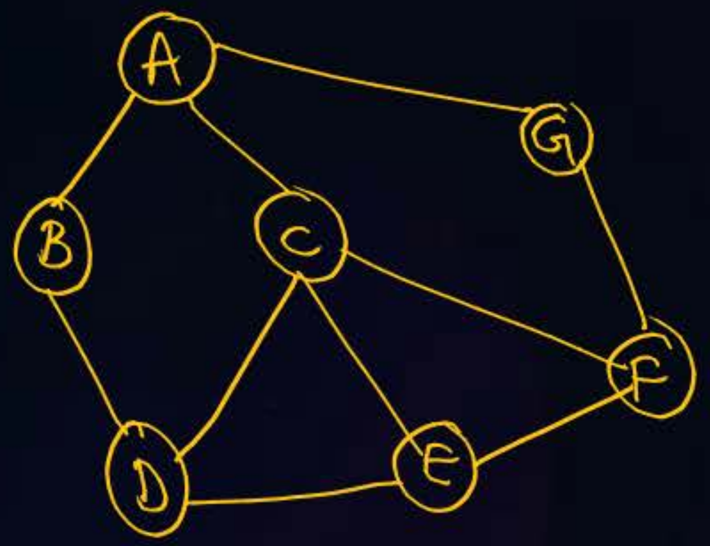
① select a Vertex V, Push on to stack S.

② select any Immediate neighbour of V and Push it.

③ Repeat ① & ② until there is no un selected vertex.

④ Then Pop a vertex until new vertices are Pushed.

Ex:



1) A G F C D B E
2) A B D E F G C
3) A C E F G D B
⋮

Valid DFS sequences

1) A B C E F G D
2) A G F C B E D
3) B D C A E F G
⋮

Invalid DFS sequences

2 mins Summary

- Graphs
  - Types of Graphs
  - Graph Traversals.

DS Syllabus Completed. 

THANK - YOU